Module 4 – Emergencies & Incidents

Train Collisions · Divided Train · Passenger Emergencies · Evacuations

Sources: Rule Book M1 (Iss 9), M3 (Iss 5), TW1 (Iss 22) · PPTs 26-01, 27-03, 28-03, 29-01

1. Train Collisions

Rule Book M1 s.1 defines a train accident as including: a collision involving trains or rail vehicles, a collision with an obstruction, a collision with a road vehicle, or a collision with a person. Sources: M1 Iss 9 s.1, M3 Iss 5 s.2, PPT 27-03.

Immediate Driver Actions (M1 s.2.1)

> *CRITICAL RULE – After any collision, switch on the hazard warning indication immediately. If not available, display a red light forward. Then make a Railway Emergency Call (REC) to the signaller.*

- Switch on hazard warning (flashing headlights) if provided; if not, display a red light forward
- Make a Railway Emergency Call (REC) to the signaller – this is the priority action
- Also tell the signaller whether the electric traction current needs to be switched off
- Stay in the cab until the signaller confirms either: the driver of each train at a stand has received the message, OR no train is approaching

Collision with an Obstruction on the Line (M3 s.2.3)

After a collision with an obstruction, bring the train to a stand and check for damage.

You must contact TOC control if any of the following could be damaged:

- Brake equipment
- Coupling and drawgear
- Vehicle suspension
- Wheelsets
- Current collection equipment
- Lifeguards
- TPWS receiver
- Any damage that may have increased the width or height of the train

TOC control will tell you one of two things:

- The train is fit to proceed (with any speed restrictions)
- The train is NOT fit to proceed until a rolling stock technician has examined it

> *CRITICAL – Being told the train is 'fit to proceed' by TOC control is NOT permission to move. You must still tell the signaller what TOC control said. The signaller tells NR Operations Control and only then authorises you to proceed.*

Detaching and Moving a Damaged Vehicle (M3 s.2.3b)

If any part of a vehicle has become loose and cannot be secured, or might make contact with the track or lineside structures:

- Get the signaller's permission before moving

- Get authority from a rolling stock technician if unsure the movement is safe
- Move passengers out of the vehicle if possible

> *Speed restriction: must not exceed 10 mph (15 km/h), or 5 mph (10 km/h) over points and crossings.*

Collisions with Other Vehicles or Buffer Stops (M3 s.2.2)

> *If any vehicle has suffered a collision or heavy impact, you must not allow it, or any part of the train, into service until a rolling stock technician has examined it. (M3 Iss 5 s.2.2)*

Person Struck by a Train (M3 + PPT 27-03)

- Refer to the 'Journey to Recovery' book — this explains what happens and the support available
- The BTP Fatality line sticker may be in the cab; some companies also have the number programmed in GSM-R
- Recognisable body parts visible only to a passing driver — movement past them can be authorised only if: excessive delay would result, remains are very close to but not foul of the running line, each driver is told the circumstances, and they agree to make the movement
- Body parts will normally be covered by police before trains are allowed to move past

2. Divided Train

Sources: M1 Iss 9 s.5, PPT 26-01. Rule Book modules M1, M3, TW1, TW7 all apply.

Recognising a Train Division

When a train becomes accidentally divided, both portions should immediately come to a stand due to brake continuity being lost. However, on certain traction types operating in degraded mode, the brake may not apply if the train divides.

Three general causes of an unsolicited brake demand:

- Operation of an in-cab safety system
- Activation of a passenger emergency alarm
- The train has become divided

> *If you cannot identify the cause of a brake application, check the tail lamp. An incomplete train will not show a tail light at the rear of the front portion.*

Immediate Actions — Establish

- The location of the divided portion
- Are all vehicles accounted for?
- Are any vehicles derailed?
- Are any other lines affected?
- Has anyone fallen from the train?
- Can the divided vehicles be re-coupled?

Contacting the Signaller

- Immediately make contact and explain the train has divided
- Advise the status of the rear portion (or that you need to walk back to locate it)
- Advise which lines need to be blocked
- Advise whether traction current needs to be isolated
- Confirm all of your train is protected by signals

> *If you think the divided portion may have derailed or is affecting other lines — make a REC.*

Driver Checklist at the Point of Division (M1 s.5.2)

- Tell passengers/on-board staff you are leaving the cab and they should remain seated
- Secure the front portion of the train
- Proceed to the point of division — secure/make safe gangway end doors on passenger stock
- Secure the rear portion of the train
- Check for any injuries requiring medical assistance
- Establish the distance between the two portions
- Inspect couplings, pipes, hoses and electrical connectors
- Check for infrastructure damage and advise signaller
- Check couplings to see if they might have damaged track/lineside equipment, or prevent recoupling (M1 s.5.2)

If the Portions CAN be Recoupled (M1 s.5.3)

- Get the personal authority of the signaller before any movement is made
- Fitters or other competent persons may be required
- After recoupling and safety checks, contact the signaller again before proceeding
- Release any brakes applied to secure the rear portion
- Carry out a brake continuity test

> *You must get the signaller's permission BEFORE any movement and AGAIN after recoupling before proceeding. (M1 Iss 9 s.5.3)*

If the Portions CANNOT be Recoupled (M1 s.5.4)

You must make sure:

- A red light is displayed at the rear of the divided portion being left behind
- A white light is displayed at the front of the divided portion being left behind

You must tell the signaller:

- That the rear portion is to be left in the section
- The exact location of the rear portion

> *You must not go beyond the next stop signal until you have told the signaller. On a single line, you must not leave the single-line section until you have told the signaller.*

Tail lamp on the front portion:

- On Track Circuit Block (TCB) lines: put a tail lamp on the rear of the front portion immediately
- On Absolute Block lines: only add the tail lamp when the front portion reaches the next signal box OR a TCB line

Other actions:

- If a passenger train — a guard or competent person should remain with the rear portion if possible
- The rear portion must remain secured at all times

Assisting Train (M1 s.5.5)

> *The assisting train driver must proceed at caution and must not exceed 25 mph (40 km/h). The signaller must tell the assisting driver the exact location and colour of light displayed on the divided portion.*

Train Protection — Lights Summary

Portion	End	Light
Rear (divided) portion	Front	White
Rear (divided) portion	Rear	Red

3. Passenger Emergencies

Sources: TW1 Iss 22 s.19, PPT 28-03.

How You Are Alerted

- Your own observation
- Activation of the emergency alarm (Passenger Communication Apparatus – PCA)
- Contact by other on-train staff
- Contact by a member of the public

PCA (Passenger Communication Apparatus) — TW1 s.19

If the PCA is operated, you must if possible avoid stopping the train in a tunnel, on a viaduct, or any other unsuitable location.

If an emergency brake application is NOT automatically applied when the alarm sounds:

- If possible, contact the person who operated the apparatus
- Ask them why the PCA has been used
- Take the necessary action
- If necessary, bring the train to a stand at a suitable location

> *CRITICAL — You MUST stop the train immediately if: you have reason to believe the train may be in*

> *danger, OR the apparatus is operated as the train is leaving a station. (TW1 Iss 22 s.19)*

> *The driver of a Driver Only (DO) train or the guard must reset the PCA before the train restarts. (TW1 Iss 22 s.19)*

Some rolling stock has an override facility. This allows you to keep the train moving while communicating via intercom with the person who operated the alarm — but the two 'must stop' conditions above always override this.

Communicating During a Passenger Emergency

- Communicate with on-train staff calmly and discreetly, especially in public view
- Passengers: communicate face-to-face where possible so they feel reassured and individual concerns can be addressed
- If using the PA system, a member of on-train staff should walk through the train to ensure everyone understood
- Communication must be clear and brief — avoid jargon or slang

If the situation is a serious emergency, contact the signaller immediately using emergency call protocols.

Information to provide to the signaller:

- The nature of the emergency
- The number and type of people affected, including severity of injury or illness
- What you require (ambulance, police, fire brigade)
- Repeat back safety-critical information to confirm understanding

Key Responders at the Scene

- Emergency Services — look for the person in the white helmet (Person in Charge of coordinating all response)
- MOM – Mobile Operations Manager
- RIO – Rail Incident Officer (Network Rail response)
- TOLO – Train Operator Liaison Officer (TOC/FOC response)

4. Evacuating a Train

Sources: M1 Iss 9 s.6, PPT 29-01. Rule Book M1 defines three types of evacuation: Controlled, Emergency, and Uncontrolled.

> *PRECONDITION: You must carry out an evacuation of a train only if it is absolutely necessary. (M1 Iss 9 s.6.1)*

The Three Types

Type	When Used
Controlled	Non-life-threatening situations. Time allows an organised, planned

	evacuation.
Emergency	Immediate danger — passengers must evacuate urgently (e.g. fire, terrorist attack). Not practicable to move to other vehicles.
Uncontrolled	Passengers self-evacuate regardless of the driver's/guard's actions — driver must still protect the line and warn passengers of risks.

Controlled Evacuation (M1 s.6.3)

EMR stranded trains policy — preferred locations in order:

- First – at a station platform
- Second – at a disused platform
- Third – end-on transfer to another train (only where both trains have end gangways)
- Fourth – side-to-side transfer to another train on an adjacent line (standard clearances required)
- Fifth – escort passengers along the track to suitable access

Driver actions — Controlled:

- Tell the signaller the train is to be evacuated and to stop any approaching train on any affected line
- If necessary, ask for the electric traction current to be switched off
- When signaller confirms no train is approaching — tell the guard

> *Ask the signaller to block all lines that will be affected and arrange for electrical switch-off/isolation before telling the guard to proceed. (PPT 29-01)*

Emergency Evacuation (M1 s.6.4)

Driver actions — Emergency:

- Carry out M1 s.2.1 actions (hazard warning on, REC to signaller)
- Tell the signaller an emergency evacuation is taking place or is necessary
- If necessary, ask for the electric traction current to be switched off
- If the signaller tells you a train is approaching — proceed on foot towards it showing a hand danger signal

> *If you cannot contact the signaller, or the signaller cannot provide signal protection, you must carry out emergency protection. (M1 Iss 9 s.6.4)*

Uncontrolled Evacuation (M1 s.6.6)

Driver actions — Uncontrolled:

- Carry out M1 s.2.1 actions (hazard warning on, REC to signaller)
- Tell the signaller an uncontrolled evacuation is taking place
- Ask for electric traction current to be switched off if necessary
- If a train is approaching — proceed towards it showing a hand danger signal

> *You must try to prevent passengers making an uncontrolled evacuation. Warn passengers who have*

> *already evacuated about risks. (M1 Iss 9 s.6.6)*

Passenger Safety (M1 s.6.5 — Guard / DO Driver)

Decide the best way to evacuate, taking into account:

- How the passengers will be moved from the site
- The need for passengers to cross the least number of lines possible to reach a safe position

> *Warn passengers to stay in a safe position until they can be escorted from the line. (M1 Iss 9 s.6.5)*

Signaller's Actions (M1 s.6.7)

- Block all lines that may be affected
- Tell the driver when protection has been provided

Marauding Terrorist Firearms Attack (MTFA) — EMR Instructions

- Make a REC via GSM-R — state 'this is an emergency call', explain the nature, block all lines, request emergency services
- Bring your train to a stand at a suitable location as quickly as possible (avoid tunnels where possible)
- Once at a stand, release train doors on the safest side
- Move yourself away from the source of danger — assist others if you can but do not place yourself in additional danger
- Keep a communication link with the signaller or control if possible (mobile phone or lineside phone)

Q&A – Test Yourself

All answers sourced from M1 Iss 9, M3 Iss 5, and TW1 Iss 22.

Train Collisions

Q1. What is your first action after any train collision?

A: Switch on the hazard warning indication (flashing headlights). If not available, display a red light forward. Then make a REC to the signaller. (M1 Iss 9 s.2.1)

Q2. Name six types of equipment damage that mean you must contact TOC control after a collision with an obstruction.

A: Brake equipment, coupling and drawgear, vehicle suspension, wheelsets, current collection equipment, lifeguards. (Also: TPWS receiver, or damage increasing width/height of train.) (M3 Iss 5 s.2.3)

Q3. TOC control tells you the train is fit to proceed after a collision. Can you move?

A: No. You must tell the signaller what TOC control said. The signaller notifies NR Operations Control. You must not move until the signaller gives you permission. (M3 Iss 5 s.2.3)

Q4. What is the maximum speed when moving a vehicle with a loose part that might contact the track or lineside structures?

A: 10 mph (15 km/h), and 5 mph (10 km/h) over points and crossings. (M3 Iss 5 s.2.3b)

Divided Train

Q5. How do you identify that a train has divided while driving?

A: Both portions should come to a stand due to brake continuity being lost. If you cannot identify the cause of a brake application, check for the tail lamp — an incomplete train will not show a tail light at the rear of the front portion. (PPT 26-01, M1)

Q6. You discover your train has divided. What is the first thing you tell the signaller?

A: That your train has become divided, the status of the rear portion if known, or that you need to walk back to locate and establish the status of the divided portion. Advise which lines need to be blocked and whether traction current needs to be isolated. (M1 Iss 9 s.5, PPT 26-01)

Q7. What lights must be displayed on the rear (divided) portion that is being left behind?

A: A red light at the rear and a white light at the front of the divided portion. (M1 Iss 9 s.5.4)

Q8. On a Track Circuit Block line, when do you put a tail lamp on the front portion after a division?

A: Immediately. On Absolute Block only add the tail lamp when the front portion reaches the next signal box or a TCB line. (M1 Iss 9 s.5.4)

Q9. What is the maximum speed for an assisting train entering the section to remove the rear portion of a divided train?

A: 25 mph (40 km/h), proceeding at caution. (M1 Iss 9 s.5.5)

Q10. After re-coupling two portions, what must you do before proceeding?

A: Contact the signaller again to confirm the train is complete and get their permission before proceeding. Also release any brakes applied and carry out a brake continuity test. (M1 Iss 9 s.5.3)

Passenger Emergencies

Q11. List four ways a driver can be alerted to a passenger emergency.

A: Own observation; activation of the emergency alarm (PCA); contact by other on-train staff; contact by a member of the public. (PPT 28-03)

Q12. The PCA is operated. The emergency brake does not automatically apply. What must you do if you cannot contact the person who operated it?

A: Take the necessary action and if necessary bring the train to a stand as soon as possible at a suitable location. (TW1 Iss 22 s.19)

Q13. In what two circumstances must you always stop the train immediately when the PCA is activated, regardless of any override facility?

A: (1) If you have reason to believe the train may be in danger; (2) if the apparatus is operated as the train is leaving a station. (TW1 Iss 22 s.19)

Q14. Who wears a white helmet at an emergency scene and what is their role?

A: The person in the white helmet is in charge of co-ordinating all response efforts (the emergency services' officer in charge). (PPT 28-03)

Evacuations

Q15. What are the three types of evacuation defined in Rule Book M1?

A: Controlled, Emergency, and Uncontrolled. (M1 Iss 9 s.6)

Q16. For a controlled evacuation, what must the driver do before telling the guard to proceed?

A: Tell the signaller the train is to be evacuated and ask them to stop any approaching train. Ask for traction current to be switched off if necessary. Wait until the signaller confirms no train is approaching, then tell the guard. (M1 Iss 9 s.6.3)

Q17. During an emergency evacuation, the signaller tells you a train is approaching. What do you do?

A: Proceed on foot towards the approaching train showing a hand danger signal. (M1 Iss 9 s.6.4)

Q18. What must you do if you cannot contact the signaller during an emergency evacuation?

A: Carry out emergency protection. (M1 Iss 9 s.6.4)

Q19. During an uncontrolled evacuation, what two things must you do regarding the passengers?

A: Try to prevent passengers making an uncontrolled evacuation, and warn passengers who have already evacuated about any risks. (M1 Iss 9 s.6.6)

Q20. What is the signaller's action when told about the evacuation of a train?

A: Block all lines that may be affected and tell the driver when protection has been provided. (M1 Iss 9 s.6.7)

Q21. List the five preferred methods for a controlled evacuation under EMR stranded trains policy, in order.

A: 1. At a station platform. 2. At a disused platform. 3. End-on transfer to another train (end gangways required). 4. Side-to-side transfer to an adjacent train. 5. Escorting passengers along the track to suitable access. (PPT 29-01)

Q22. You are involved in a train collision. When must you stay in the cab rather than leaving it?

A: You must stay in the cab until the signaller tells you either: the driver of each train that must remain at a stand has received and understood the message, OR no train is approaching. (M1 Iss 9 s.2.1)